

ENGINEERING PORTFOLIO

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TABLE OF CONTENTS

01

NASA HERC Rover

Suspension system design, FEA analysis, fabrication planning, and competition performance for a human-powered extraterrestrial rover.

02

Advanced CAD Drawings

42-drawing recreation of complex mechanical components and assemblies in Autodesk Inventor, from 2D documentation to manufacturing-ready 3D models.

03

Machine Guarding Redesign

Industrial guarding redesign improving maintainability, impact resistance, and installation speed for weekly maintenance crews.

04

High-Powered Rocket

Aerodynamic design, flight simulation, and successful launch of a high-powered rocket.

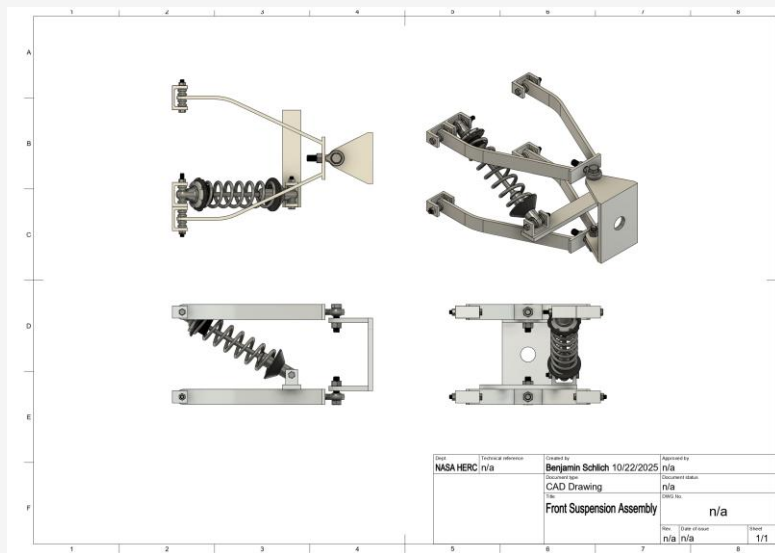
05

Robot Camera Mount

Custom SOLIDWORKS-designed camera mount integrating an Intel RealSense D435i onto a UR5e robotic arm for vision-based object detection.

01

NASA HERC Rover



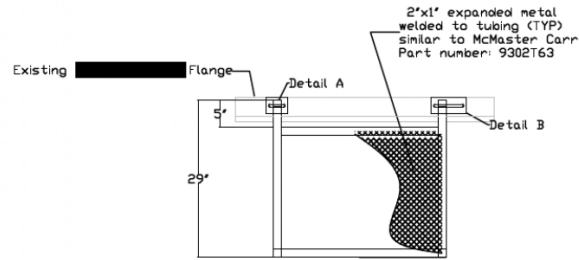
[Front suspension.mp4](#)



Rear suspension

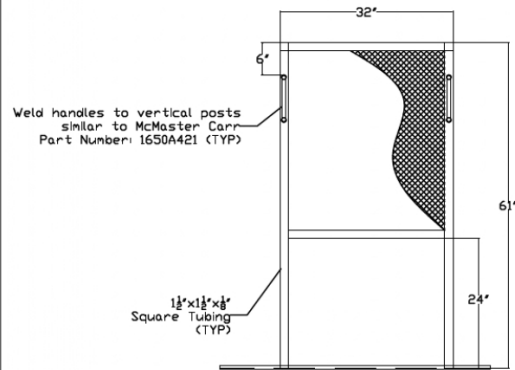
02

Advanced CAD Drawings.pdf



1
SK-1

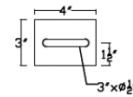
V1 Guards—Top View
N.T.S.



2
SK-1

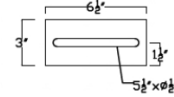
V1 Guards—Front View
N.T.S.

Detail A:
Single clip
channel bracket



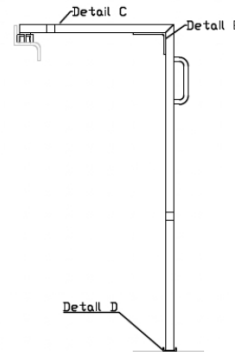
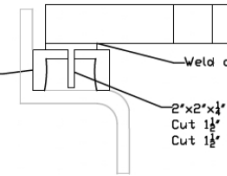
Each section should not exceed 65 pounds

Detail B:
Double clip
channel bracket



Detail C:
Clip and channel
connection point

C3x6 steel channel
Weld to existing Flange
McMaster Carr
Part Number: 7779T57



Detail E:
6"x4"x5/16" Angle Iron Support
Cut 1½" Wide
McMaster Carr
Part Number: 6983N29



Detail D:
Channel slot for
legs



2½"x8"x½" (Outside Dimensions)
U-channel McMaster Carr
Part Number: 7779T101

3
SK-1

V1 Guards—Side View
N.T.S.

Guards

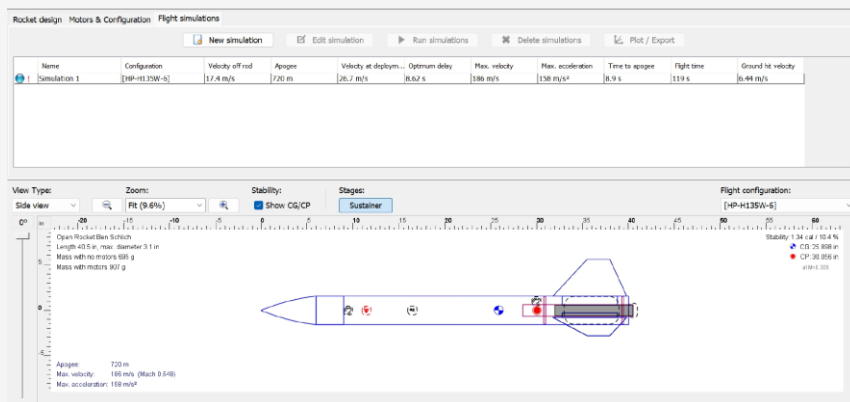
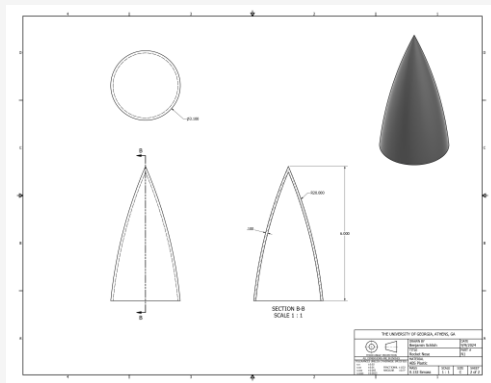
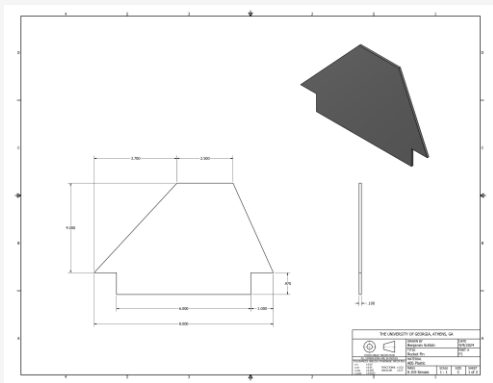
V1

NO DESCRIPTION DATE
REVISION
SEAL

DRAWN BY
DESIGNED BY
CHECKED BY
USE NO
SCALE
DATE
SHEET TITLE

SHEET NO.

05 High-Powered Rocket



05 Robot Camera Mount

